Noa Moriel **Curriculum Vitæ**

CONTACT INFORMATION

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EDUCATION

Hebrew University, Jerusalem, Israel. October 2020 – Ongoing
Ph.D. Computer Science.
Advised by Dr. Mor Nitzan.

Hebrew University, Jerusalem, Israel. September 2016 – December 2018
M.Sc. Computer Science and Computational Biology. **(GPA 95.10/100)**
Finished 2nd in class.
Advised by Prof. Nir Friedman.

University of Toronto, Toronto, Canada. September 2015 – January 2016
M.Sc. Computer Science exchange program.

Hebrew University, Jerusalem, Israel. September 2012 – August 2015
B.Sc. Computer Science and Computational Biology. **(GPA 94.25/100)**
Magna Cum Laude.
Advised by Prof. Nir Friedman.

WORK EXPERIENCE

Harvard University, Postgraduate Researcher. September 2019 – September 2020
Reconstructing spatial information from single-cell RNA-Seq data.
Advised by Dr. Mor Nitzan.

Mobileye, Algorithm Engineer. March 2018 – August 2019
Advancing object detection of Mobileye's autonomous vehicles.

Broad Institute, Graduate Researcher. February 2016 – June 2016
Adopting single-cell RNA-Seq methods to yeast, both technologically and computationally.
Designing a libarary of combinatorial genetic perturbations.
Advised by Prof. Nir Friedman, while working at Prof. Aviv Regev's lab.

PUBLICATIONS

Moriel N.*, Ricci M.*, & Nitzan M. 2024. Let's do the time-warp-attend: learning topological invariants of dynamical systems. ICLR 2024.

Moriel N.*, Edvin M.*, & Nitzan M. 2024. Optimal sequencing budget allocation for trajectory reconstruction of single cells. Submitted to ISMB 2024.

Ricci M., **Moriel N.**, Piran Z., & Nitzan M. 2023. Phase2vec: dynamical systems embedding with a physics-informed convolutional network. ICLR 2023.

Adler M.*, **Moriel N.***, Goeva A.*, ..., Regev A., Medzhitov R., & Nitzan M. 2023. Emergence of division of labor in tissues through cell interactions and spatial cues Cell Reports.

Mages S.*, **Moriel N.***, Avraham-Davidi I., ..., Klughammer J., Regev A., & Nitzan M. 2023. TACCO: Unified annotation transfer and decomposition of cell identities for single-cell and spatial omics. Nature Biotechnology.

Avraham-Davidi I., Mages S., Klughammer J., **Moriel N.**, ..., Roper J. ,Rozenblatt-Rosen O., Nitzan M., & Regev A. 2022. Integrative single cell and spatial transcriptomics of colorectal cancer reveals multicellular functional units that support tumor progression. *Under review* at Nature Cancer. bioRxiv.

Vaknin Y., **Moriel N.**, & Nitzan M. 2022. Using co-localization priors and microenvironment statistics to reconstruct tissue organization from single-cell data. NeurIPS LMRL Workshop 2022.

Moriel N.*, Senel E.*, Friedman N., Rajewsky N., Karaiskos N. & Nitzan M. 2021. NovoSpaRc: Flexible Spatial Reconstruction of Single-Cell Gene Expression with Optimal Transport. Nature Protocols.

* Equal contribution

Teaching

Supervising assistant of Computational Methods in Molecular Biology Lab.	2024
Teaching assistant of Introduction to Machine Learning.	2023
Teaching assistant of Introduction to Artifical Intelligence.	2022
Teaching assistant of Genomics.	2017
Teaching assistant of Algorithms in Computational Biology.	2017

Awards and Excellence Programs

Council of Higher Education PhD fellowship.	2021/2022
CIDR PhD fellowship in core data science.	2021/2022
CIDR PhD fellowship in core data science.	2020/2021
Participated in Weizmann's excellence program, Ulpanot De-Shalit.	2014
The Hebrew University Dean's Award.	2012/2013
The Hebrew University Dean's Award.	2013/2014
IDF (Israeli Defense Force) Intelligence Excellence Award.	2010
Participated in Weizmann's excellence program, ISSI.	2010
Participated in Microsoft's excellence program, Hadarim.	2010
Participated in young diplomats program, Shagririm Tzehirim.	2010

Military service

Pilot training, served in 8200.	2010 – 2012